

FedPot: A Quality-Aware Collaborative and Incentivized Honeypot-Based Detector for Smart Grid Networks

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Abstract—Honeypot technologies provide an effective defense strategy for the Industrial Internet of Things (IIoT), particularly in enhancing the Advanced Metering Infrastructure's (AMI) security by bolstering the network intrusion detection system. For this security paradigm to be fully realized, it necessitates the active participation of small-scale power suppliers (SPSs) in implementing honeypots and engaging in collaborative data sharing with traditional power retailers (TPRs). To motivate this interaction, TPRs incentivize data sharing with tangible rewards. However, without access to an SPS's confidential data, it is daunting for TPRs to validate shared data, thereby risking SPSs' privacy and increasing sharing costs due to voluminous honeypot logs. These challenges can be resolved by utilizing Federated Learning (FL), a distributed machine learning (ML) technique that allows for model training without data relocation. However, the conventional FL algorithm lacks the requisite functionality for both the security defense model and the rewards system of the AMI network. This work presents two solutions: first, an enhanced and cost-efficient FedAvg algorithm incorporating a novel data quality measure, and second, FedPot, the development of an effective security model with a fair incentives mechanism under an FL architecture. Accordingly, SPSs are limited to sharing the ML model they learn after efficiently measuring their local data quality, whereas TPRs can verify the participants' uploaded models and fairly compensate each participant for their contributions through rewards. Moreover, the proposed scheme addresses the problem of harmful participants who share subpar models while claiming high-quality data through a two-step verification approach. Simulation results, drawn from realistic mircorgrid network log datasets, demonstrate that the proposed solutions outperform state-of-the-art techniques by enhancing the security model and guaranteeing fair reward distributions.

Index Terms—AMI, honeypot-based detector, security model, machine learning, incentive mechanism, collaborative learning.

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NOMENCLATURE

$V(z)$	global model at z -th training round
V_m	local model of SPS m
$V_m(j)$	the local model update at j local iteration
η	the learning rate
$\Lambda(x_i, \delta)$	an open ball space with a radius of δ centered at x_i
$\mathcal{S}_z$	the selection set at round z
π_m	the model and its rewards of type m
θ_m	SPS type: $\theta_1 \leq \dots \leq \theta_m \leq \dots \leq \theta_M$
$\varphi(\mathcal{D}_m)$	the estimation of the local VDD quality
ξ	the defense model size
C_d	the honeypot deploying cost
C_m	the local deployment, training, and uploading costs
C_t	the local training cost to update the defense model
C_u	the uploading cost of the local defense model
E	number of local training epochs
$F_m(V_m(j))$	the local loss function at j local iteration
$F_m(V_m)$	the updated local loss function
$G(V_m)$	the revenue given by each SPS m
M	number of connected SPSs
R_m	the reward given to SPS m
S_z^m	the selection binary variable
T^{max}	the maximum latency in each round
U_m	the utility of SPS m
U_{TPR}	the utility of TPR (i.e., utility company)
Π	Incentive set for all types
$F_i(\mathbf{V}_m, x_i, y_i)$	loss function on sample i
AMI	Advanced Metering Infrastructure
CPU	Central Processing Unit
DDoS	Distributed Denial-of-Service
DL	Deep Learning
DoS	Denial-of-Service
FL	Federated Learning
GDPR	General Data Protection Regulation
IC	Incentive Compatibility
IID	Independent and Identically Distributed
IIoT	Industrial Internet of Things
IR	Individual Rationality
LDIC	Local Downward Incentive Compatibility
LUIC	Local Upward Incentive Compatibility

MITM	Man-in-the-Middle
ML	Machine Learning
NIDS	Network Intrusion Detection System
PLCs	Programmable Logic Controllers
RTUs	Remote Terminal Units
SG	Smart Grid
SPSs	Small-Scale Power Suppliers
TNR	True Negative Rate
TPRate	True Positive Rate
TPRs	Traditional power retailers
VDD	Valid Defense Data

I. INTRODUCTION

A SMART Grid (SG) is an advanced electricity system that leverages digital tech, IIoT, and networking to boost efficiency, reliability, and sustainability. It achieves this through a cyber-physical system for the bidirectional flow of power and information, automating supply-demand balance with real-time data [2]. SG allows individuals to become small-scale power suppliers (SPSs) using renewable sources, reducing the burden on traditional power retailers (TPRs) and fostering advanced metering infrastructure (AMI) integration [3], [4].

However, integrating different network technologies to interconnect SPSs and TPRs, which lack a proper defense system, is raising concerns about the security and privacy of SG systems. Adversaries have access to numerous vulnerabilities, including ways to launch destructive attacks and access sensitive information such as a homeowner's residential address, social security number, daily habits, and any information related to unauthorized electricity consumption or disruption in the network. As an example of such attacks, the 2015 Ukrainian power outage illustrates the critical vulnerabilities to cyber-attacks in both the control center and the smart devices employed for managing and observing the electrical system [5].

In addition, SG is subject to denial-of-service (DoS) attacks, which can flood the network with traffic, causing delays in data transmission and processing. This can lead to disruptions in the system's normal operation and potentially cause critical elements of the energy system to fail [6]. To protect the communication infrastructure, implementing a Network Intrusion Detection System (NIDS) is crucial. NIDS serves as a robust shield, detecting numerous threats that extend beyond the capacity of traditional firewalls. The protective capability of NIDS has been notably enhanced by the advent of Machine Learning (ML) and Deep Learning (DL)-based approaches, pushing their performance to a significantly higher level. This interconnected relationship between NIDS, ML, and DL demonstrates a synergistic blend of network security and advanced computational methodologies [7].

Building upon this security paradigm, incorporating honeypot technology further enriches the protective capabilities of ML/DL-based NIDSs. Honeypots empower these systems to meticulously map the attack surface, discern patterns, and thwart malicious actions by delivering detailed insights into potential intruder behavior. Within the realm of SG, a honeypot

mimics the regular operations of a meter with the intent to deceive, misdirect, and analyze the activities of potential intruders. Employing such tactics allows SPSs to develop independent, streamlined security measures. Furthermore, it facilitates the exchange of defensive information with TPRs, eliminating the need for TPRs to purchase costly security models from security retailers. This approach amplifies the protective layers of AMI, creating a more robust defense mechanism. Simultaneously, it alleviates the financial strain associated with defense strategies for TPRs, presenting a dual advantage by strengthening network security while managing costs effectively [8].

Significant effort in the literature has been devoted to developing incentive mechanisms and designing contracts to encourage SPSs to implement honeypots and share protective information with TPRs while maintaining a balance such that SPSs do not reap excessive benefits beyond their due. For example, in [9], [10], the authors proposed information asymmetry-based contract theory approaches considering different communication systems. Concentrating on the SG network, the work in [11] introduced a recent motivation-driven approach where TPR motivates SPSs to deploy honeypots and exchange defensive information to enhance the security framework. This approach relies on a range of essential prerequisites, including the dimensions of the shared data and the related expenses. However, all the prior work may violate the SPSs' privacy while increasing the cost of sharing the obtained raw data. The recent enforcement of stringent data privacy regulations, such as the General Data Protection Regulation (GDPR) [12], further underscores the importance of maintaining data privacy. Additionally, the large volume of honeypot logs can lead to excessive transmission costs and network congestion. It is also worth noting that the shared honeypot logs may not always yield benefits, as they could contain redundant information that does not enhance the existing protection strategy. Given the aforementioned concerns, federated learning (FL) [13] has become an effective distributed ML/DL method to maintain privacy and minimize communication costs by exclusively sharing the ML model while retaining the raw data in its original repository. This seamlessly connects the need for privacy preservation and cost-efficiency with the advantages of shared data for network security. Current research on FL typically assumes that entities are willing to participate in the FL training process and use the collected data to improve the shared model [14], [15]. In reality, entities may be reluctant to join without properly designed incentives (i.e., contracts) because FL consumes significant resources (i.e., computation and communication costs). Furthermore, entities in FL are autonomous actors who decide when and how to interact. When dealing with different reward strategies from different alliances, participants may use different training techniques, influencing the performance of the resulting models. To that end, it is critical to develop an efficient reward mechanism to encourage entities to participate in FL while maintaining the required level of data quality. Researchers have recently presented several incentive schemes aimed at compensating the involved parties (i.e., SPSs) using financial incentives according to the magnitude of their data

contributions [16], [17], [18]. Utilizing current FL methodologies, all participants acquire the same model upon the end of the training phase, regardless of the produced data quality (i.e., honeypot logs) or the impact of the submitted local models. Consequently, certain participants possessing large datasets may provide inadequate contributions yet attain a higher proficiency level than others who possess high-quality data. This scenario gives rise to a challenging issue called the FL free-rider problem.

A. Contribution

Considering the previously mentioned remarks, we introduce FedPot, an FL framework initially designed for SG networks but validated across diverse scenarios, including IEC 104 and N-BaIoT datasets. This architecture incorporates refined, efficient, and resilient aggregation and averaging techniques complemented by a fair rewards system based on data quality. While our primary case study is centered on SGs, our approach of combining honeypot logs and FL has shown adaptability and effectiveness across multiple domains. We introduce novel schemes for local data quality, participant selection, and global model averaging, where the TPR resolves a convex optimization problem that prioritizes data quality over data size. Each SPS fine-tunes the global model received from the TPR using its honeypot logs and transmits back the model parameters. After that, the TPR combines and enhances the defensive model employing the proposed approaches, as detailed later. To overcome the FL free-rider problem within AMI networks, we propose a novel metric to measure local data quality and contributions instead of data size, which may contain redundant information that does not improve the security defense model. We devised a two-step verification process to tackle malicious or poorly performing participants. The TPR verifies submitted models, then updates the global model and allocates incentives based on contribution. In summary, the primary contributions of this work are:

- Develop an effective architecture for privacy-preserving honeypot-based detectors, FedPot, that protects user privacy while considering data quality, an efficient global model, and fair incentives. The proposed solution handles and ensures a reliable FL training process based on valid defense data acquired by implementing honeypots on the SPS side.
- Formulate the problem as an optimization problem, then provide solutions incorporating (i) problem reformulation and transformation, (ii) the prior quality determination of the local data through novel metrics, and (iii) two schemes for reliable global averaging and contribution-based reward distributions.
- In response to the challenge of adversarial perturbations in model uploads, we introduce a universally applicable two-step verification system. This robust approach is designed to ensure the integrity of model contributions.
- Carry out comprehensive simulations using realistic log data from various datasets (i.e., N-BaIoT, IEC 104, and IEC MMS datasets). The results affirm that our proposed framework outperforms existing

state-of-the-art techniques in multiple application domains.

B. Organization

The remainder of the paper is structured as follows. Section II briefly reviews the relevant literature on NIDSs and honeypot deployment. The system models, including the learning, cost, and reward, are presented in Section III. The problem is formulated in Section IV, and the proposed solution, including the problem reformulation and transformation, is presented in Section V. In Section VI, we discuss the experimental setup and present the numerical results. Finally, Section VII concludes this paper and suggests possible directions for further research.

II. RELATED WORK

With the proliferation of microgrids, major concerns about cyberattacks on such systems via smart meters have arisen. The United States Department of Homeland Security reported 224 destructive cyber intrusions against local electric utilities between 2013 and 2014 [19].

A. AMI Security

Security concerns have been intensively investigated in the past few years as a key component of the IIoT. Authors in [20] studied the security approach to mitigate cyber-attacks in the context of the IIoT. The main assumption was that the attackers have sufficient tools to identify advanced vulnerabilities that enable them to attack IIoT systems. In [21], Li et al. used consortium blockchain technology to overcome transaction constraints in the IIoT. However, fewer studies have been conducted on the security of various components of AMI systems. The work in [22] introduced a security protocol that preserves AMI private information while securely delivering control packets at the exact time. In [23], Fasial et al. investigated the feasibility of employing data stream mining to improve AMI cybersecurity via NIDS. Yan et al. [24] describe an SG AMI security framework where different security concerns associated with AMI deployment are considered.

B. ML/DL-Based and FL-Based NIDSs

Many ML algorithms have been utilized to boost the NIDS in the past years to understand complicated network traffic better [25], [26], [27]. ML/DL-based NIDSs are used to identify unknown intrusions by analyzing the statistical characteristics of the network traffic. However, DL-based solutions have shown better performance, especially in extracting knowledge from complicated features rather than the shallow features in ML-based detectors [25], [26], [27]. Recently, FL has been increasingly employed in the realm of NIDSs for the collaborative design of ML and DL-based detectors [28]. Specifically, the work presented in [29] developed a cooperative detector capable of identifying zero-day botnet attacks in oT networks using FL. Extending FL's utility to energy systems, [30] introduced a privacy-preserving and communication-efficient FL-based energy predictor for net-metering systems. This approach combines a hybrid DL

denotes the local loss function, quantifying the model's error concerning local data samples, D_m signifies the quantity of data examples in the gathered records (i.e., traffic logs), and η stands for the learning rate. Upon completion of local training by all chosen SPSs, their updates are transmitted to the TPR. Then, the TPR consolidates these updates and modifies the global model given by:

$$\mathbf{V}(\mathbf{z}) = \sum_{m=1}^M \frac{D_m}{D} \mathbf{V}_m, \quad (3)$$

where $D = \sum_{m=1}^M D_m$, represents the aggregate samples across all SPSs. Moreover, to guarantee that the attained accuracy is generalized, the TPR evaluates the uploaded models and provides incentives only to SPSs exceeding the threshold, ψ_m , as determined by the quality assessment algorithm discussed subsequently. SPSs that pass the quality evaluation receive rewards. The TPR iterates this procedure for Z global iterations until the shared security model reaches the target accuracy level. In this paper, we highlight that the incentive set is constructed every global round, where the TPR aims to sustain the desired accuracy while allowing for a slight increment in succeeding rounds.

A. Participation Willingness

In addition to the cost of deploying the honeypot, the suppliers bear the cost of training and uploading the local models. This cost includes the time and energy consumption for the honeypot deployment as well as training and uploading their local models. As such, the TPR cannot serve all existing devices due to insufficient bandwidth sub-channels. Thus, in every FL round, just a subset of the participants $\mathcal{S}_z$ can send their updates. The chosen set is described as follows:

$$\mathcal{S}_z = \{m \mid S_z^m = 1, m = 1, 2, \dots, M\}, \quad (4)$$

where $S_z^m = 1$ denotes that device m is in the selected set $\mathcal{S}_z$, otherwise $S_z^m = 0$. The latency among the chosen set consists of two parts: uploading latency and computing latency. For the uploading latency, all devices have the same defense model structure determined by TPR, with a size v_z denoted by ξ . Generally, we consider that the orthogonal frequency division multiple access (OFDMA) scheme is used for the communication between the BS and the devices, where each m -th device is given bandwidth of size $\lambda_z^m B$. Additionally, based on the size of the model, the system bandwidth is split into N sub-channels given by: $N = \frac{B}{\xi}$, where each participant is assigned an l -th sub-channel of size $\lambda_z^m B$. As a result, the m -th device is capable of achieving a data rate of $\alpha_z^m = \lambda_z^m B \ln(1 + \frac{P_z^m |h_z^m|^2}{N_0})$, where h_z^m represents the channel gain between the BS and client m , while P_z^m denotes the transmit power of the m -th client, and N_0 denotes background noise (i.e., Additive White Gaussian Noise (AWGN)). Hence, the uploading latency is given by: $T_m^{trans} = \frac{\xi}{\alpha_z^m}$, and the related transmission cost is defined as: $C_u = T_m^{trans} p_m^{trans}$, where p_m^{trans} is the transmit power. Regarding the calculated latency, every device requires a time duration defined as: $T_{cmp}^m = E \frac{\phi_m D_m}{f_m}$, to train its local model, where ϕ_m (cycles/sample)

denotes the number of processing cycles to execute one sample, and f_m (cycles/second) is the central processing unit (CPU) frequency. Accordingly, the training cost is defined as: $C_t = \kappa_m f_m^3 T_{cmp}^m$, where κ_m is the capacitor's coefficient related to the chip. As a result, the combined latency for computation and uploading for each m -th participant during the z -th round is expressed as: $T_{total}^m = T_{trans}^m + T_{cmp}^m$. In a real FL scenario, the coordinating server (i.e., TPR) establishes a time constraint, i.e., round deadline constraint, ensuring that every SPS participant completes their tasks within this specified timeframe. Specifically, this time can be identified based on the latency of the slowest chosen SPS $m \in \mathcal{S}_z$, which is defined as:

$$T^{max} = \max\{S_z^m (T_{trans}^m + T_{cmp}^m)\}. \quad (5)$$

It is worth emphasizing that while SPSs do benefit from the globally trained model, this alone does not ensure a model tailored to their unique threat landscape. Incentives encourage SPSs to actively contribute quality data, keeping the model updated and effective against evolving cybersecurity threats, leading to solving the free-rider problem in FL. Incentives also offset the operational costs of maintaining honeypots and promote diverse participation. This results in a more robust and tailored global model, rewarding contributors with a system better suited to their security needs.

B. SPS and TPR Utilities

To calculate the SPS utility, which involves implementing the honeypot and exchanging information associated with the VDD data (i.e., the updated local model), we initially have to compute the total incurred cost in the following manner:

$$C_m = C_d + C_t + C_u, \quad (6)$$

where C_d represents the cost related to running the honeypot, C_t is the cost related to updating the local model utilizing the gathered logs, and C_u is the cost related to uploading the local update. Accordingly, the utility of each SPS is expressed as:

$$U_m = \ln(\theta_m R_m) - C_m. \quad (7)$$

Equ. (7) implies that while the reward increases, the utility does not increase at the same rate due to the diminishing returns property of the logarithmic function. Once all SPSs taking part in the updating and uploading of the global defense model finish their tasks, the utility of the TPR is computed as follows:

$$U_{TPR} = \sum_{m=1}^M (\theta_m G(V_m) - R_m), \quad (8)$$

where $G(V_m)$ represents the revenue obtained from each SPS's update. Specifically, each SPS shares its model, V_m , which will add a profit, $G(V_m)$ to TPR with associated costs.

C. Adversarial Model and Assumption

In this research paper, we analyze two stages of adversarial behavior using the N-BaIoT dataset. The first stage involves deploying botnets to carry out DDoS attacks by introducing

networked zombies into the SPSs. The SPS implements a honeypot system to counter these threats, collect valuable information about the attack surface, and document all activities. These logs are then integrated into a collaborative model training approach, enabling information exchange with the TPR and enhancing understanding of the attacker's behavior. The second stage of adversarial conduct occurs during the FL process when a malicious SPS falsely claims superior local data quality while contributing subpar updates to the model. To develop a secure and efficient FL process that can mitigate such issues, we put forward two distinct aggregation and averaging techniques. These methods will be discussed in greater detail later in the paper, providing insight into their practical applications and advantages.

IV. PROBLEM FORMULATION

The TPR aims to optimize data quality to boost the performance of the target model and assures fair incentives to all participating SPSs. This is equivalent to the optimization problem given by:

$$\mathbf{P1.} \quad \max_{\Pi} : \sum_{z=1}^Z \sum_{m=1}^M S_z^m (\theta_m G(V_m) - R_m) \quad (9)$$

such that

$$C1: \ln(\theta_m R_m) - C_m \geq 0, \forall m \in \{1, 2, \dots, M\}, \quad (10)$$

$$C2: \ln(\theta_m R_m) - C_m \geq \ln(\theta_m R_{m'}) - C_{m'}, \quad (11)$$

$$C3: \sum_{z=1}^Z \sum_{m=1}^M R_m \leq B, \quad (12)$$

$$C4: T_m^{total} < T^{max}, \quad \forall m \in \{1, 2, \dots, M\}, \quad (13)$$

$$C5: R_m > 0, \quad (14)$$

where $C_m^{m'}$ is the cost of the supplier in type m , selecting a contract of type m' , and B is the allocated total budget for upgrading the security defense model. In (9), C1 is the individual rationality (IR) constraint in which each supplier has to gain non-negative utility. Constraint C2 is incentive compatibility (IC), in which each supplier should choose the exact incentive aligned with its type. In C3, the retailer guarantees that the rewards delivered to the participants do not surpass the allocated budget. The delay constraint given by C4 ensures that participants complete the assigned tasks in a defined period. In C5, each participating SPS should receive non-negative rewards. It is worth noting that the TPR can pay for the model upgrade from the security market if the required rewards are high and exceed its allocated budget.

Finding the direct solution for (9) is extremely intricate due to the need for previous knowledge of all participants over all rounds, which is nearly unattainable. Additionally, the contracts must be adjusted based on the SPSs' participation over the training period since contributions decrease over time, and the model may converge to various stationary points, resulting in slower convergence. To address these challenges, we propose to transform the problem into an online problem

in which the retailer can redesign the contracts every round. The problem is reformulated as:

$$\mathbf{P2.} \quad \max_{\Pi} : \sum_{m=1}^M S_z^m (\theta_m G(V_m) - R_m) \quad (15)$$

such that

$$C1: \ln(\theta_m R_m) - C_m \geq 0, \forall m \in \{1, \dots, M\}, \quad (16)$$

$$C2: \ln(\theta_m R_m) - C_m \geq \ln(\theta_m R_{m'}) - C_{m'}, \quad (17)$$

$$C3: \sum_{m=1}^M R_m \leq B_z, \quad (18)$$

$$C4: T_m^{total} < T^{max} \quad \forall m \in \{1, 2, \dots, M\}, \quad (19)$$

$$C5: R_m > 0, \quad (20)$$

where B_z is the budget allocated for round z . For example, $B_z = \frac{B}{Z}$, if we aim to allocate a fixed budget in each round, it can be adjusted depending on the TPR's gain.

Clearly, (15) is intractable, and the constraints are coupled. In particular, constraint C1 in (15) related to IR and $M(M-1)$ constraints related to IC make finding a direct solution for (15) very challenging. Thus, we start by reducing the IR and IC constraints. Then, we propose FedPot, a framework that includes tractable solutions to these challenges. FedPot is divided into two blocks, one on the SPS side and one on the TPR side. On the TPR side, we initially use the local data quality to ensure maximum utility for the TPR. Then we relax the problem to enable the TPR to select the optimal participating SPS. Next, we propose two averaging schemes considering the security aspects. To ensure fair incentives, we also proposed two reward schemes based on the claimed data quality and the contribution to the global model.

V. PROPOSED SOLUTION

This section introduces the proposed solutions, including how we reduced the constraints' complexity, the prior quality determination of the local VDD, the trusted and untrusted model averaging schemes, and the rewards distribution. The proposed solution for the SPS side starts by deploying a honeypot on the SPS side to record all network traffic into log files. The logs are then transformed into a readable format (i.e., CSV) and cleaned with an extensive preprocessing step. It is worth mentioning that the practice of anonymizing data (i.e., CSV files) is a commonly used method to protect sensitive information. However, several factors motivate us to use FL rather than simply anonymizing the data. First, FL mitigates the risk of data re-identification, a vulnerability in anonymization. Second, unlike anonymized CSV files, FL only shares aggregated model parameters, preserving data patterns while reducing exposure risks. Third, FL minimizes the chance of large-scale breaches by avoiding centralized data storage. Lastly, FL enables collaborative learning across distributed honeypots without centralization, reducing communication and storage costs. We propose a local evaluation scheme in which each SPS evaluates its local VDD to determine whether to participate in the FL learning process. It is important

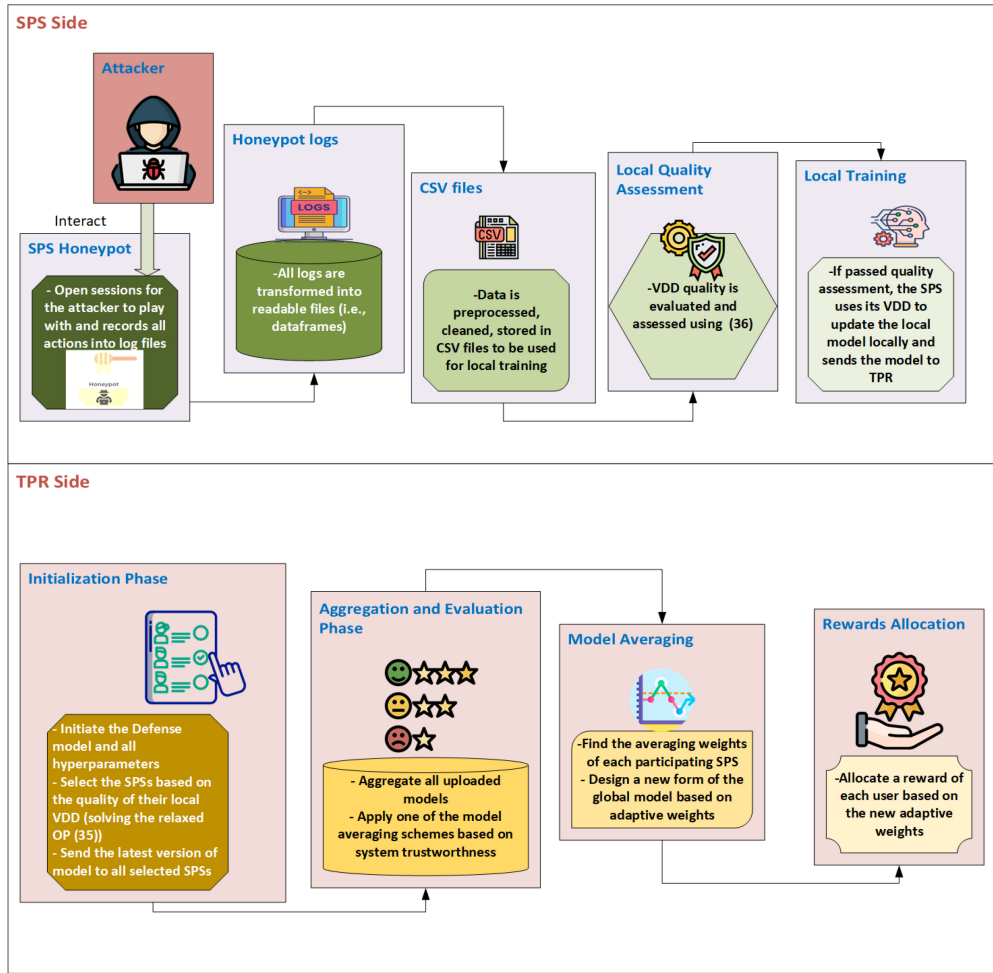


Fig. 2. FedPot Architecture.

to mention that the SPS is willing to participate only if the required quality is achieved. Fig. 2 illustrates the whole procedure performed at both the SPS and TPR.

A. Constraints Complexity Reduction

As stated previously, it is almost intractable to directly solve (15). To simplify, we present the subsequent lemmas.

Lemma 1: Given the budget for each round B_z , for any feasible contract (V_m, R_m) , $R_m \geq R_{m'} \iff m \geq m' \forall m, m' \in \{1, 2, \dots, M\}$.

Proof: We first prove that if $\theta_m \geq \theta_{m'}$ where $m \geq m'$, then $R_m \geq R_{m'}$. Adding the IC constraints for both types θ_m and $\theta_{m'}$, yields:

$$\ln(\theta_m R_m) - C_m \geq \ln(\theta_m R_{m'}) - C_{m'}, \quad (21)$$

and

$$\ln(\theta_{m'} R_{m'}) - C_{m'} \geq \ln(\theta_{m'} R_m) - C_m, \quad (22)$$

We add both (21) and (22), we have:

$$\ln(\theta_m - \theta_{m'})(R_m - R_{m'}) \geq 0. \quad (23)$$

Thus, $R_m \geq R_{m'}$. ■

From Lemma 1, we note that more rewards will be given to more participants. This means that Lemma 1 is monotonic.

Hence, this analysis of IC constraints reduces the IR constraints as indicated in the following lemma.

Lemma 2: Given the B_z , and Lemma 1 related to IC constraints and the sorted participants based on their data quality, the IR condition can be reduced as:

$$\ln(\theta_1 R_1) - C_1 \geq 0 \quad (24)$$

Proof: Given the sorted SPSs based on their types as defined in Section III, the IC constraints are used as follows:

$$\ln(\theta_m R_m) - C_m \geq \ln(\theta_m R_1) - C_1 \geq \ln(\theta_1 R_1) - C_1 \geq 0. \quad (25)$$

From Section V-A, we note that if the first SPS's type meets the IR constraint, all other SPSs' types will automatically meet other IR constraints. Therefore, the IR of type 1 is sufficient to achieve all other IR constraints. ■

According to Lemma 1, Lemma 2, we can transform the IC constraint into local downward incentive compatibility (LDIC) as follows [34]:

$$\begin{aligned} \ln(\theta_m R_m) - C_m &\geq \ln(\theta_m R_{m-1}) - C_{m-1}, \\ \forall n &\in \{2, 3, \dots, M\}, \end{aligned} \quad (26)$$

and local upward incentive compatibility (LUIC) as follows:

$$\begin{aligned} \ln(\theta_m R_m) - C_m &\leq \ln(\theta_m R_{m+1}) - C_{m+1}, \\ \forall n &\in \{1, 2, \dots, M-1\}. \end{aligned} \quad (27)$$

From Equations (26), (27), we note that participants should be given rewards based solely on their contributions. Moreover, we observe that the objective function in (15) is decreasing w.r.t R_m and increasing w.r.t $G(V_m)$.

B. Quality Measure of the SPSs' VDD

The SPSs asymmetrically exchange information, sharing only the local model updated using the collected VDD with the TPR, rather than the VDD itself. Hence, the TPR is necessary to validate the uploaded models. The quality of the local model depends on several factors, including the variety of the data, the number of contained classes, and the number of updates performed. In this regard, we first model the SPS's data by considering the data quality, and then by assessing the impact of the local models on the generalization of the global model. This strategy allows the TPR to guarantee a robust model even in the presence of malicious participants. Each local dataset $\mathcal{D}_m$ generally comprises traffic data with input-output pairs of $\{\mathbf{x}_{i,d}^{(m)}, y_i^{(m)}\}_{i=1}^{D_m}$, where $\mathbf{x}_{i,d}^{(m)} \in \mathbb{R}^d$ denotes the input holding d attributes, and $y_i^{(m)} \in \mathbb{R}$ denotes the matching class label. Given that every SPS has its own network traffic patterns and a varying response according to its activity, the honeypot logs produce data of various sizes and follow a non-identical and independently distributed (non-i.i.d.) manner. Furthermore, the logs in each SPS may have distinct attack types, and some of the SPSs might not have any malicious samples. Let $\mathcal{D} = \{\mathcal{D}_1 \cup \mathcal{D}_2 \cup \dots \cup \mathcal{D}_M\}$ represent the aggregated dataset from all SPSs. We define the probability of a sample, i , being included in the local logs as:

$$\rho(\mathcal{D}_m|i) = \begin{cases} 1 & : i \in \mathcal{D}_m \\ 0 & : \text{otherwise} \end{cases} \quad (28)$$

Definition 1 (δ -Data Coverage): With a predetermined radius, the coverage of data collection through sampling is obtained by:

$$\rho(\mathcal{D}_m, \delta) = \mathcal{D} \cap \cup_{x_i \in \mathcal{D}_m} \Lambda(x_i, \delta), \quad (29)$$

where $\Lambda(x_i, \delta)$ is an open ball space with a radius of δ centered at x_i .

Considering the data space among all SPSs is an Euclidean space, the range of δ is confined to the interval $[0, \sqrt{d}]$.

Definition 2 (VDD Quality): $\varphi(\mathcal{D}_m)$ gives the estimation of the local VDD quality measurements defined as:

$$\varphi(\mathcal{D}_m) = \frac{1}{\sqrt{d}} \int_0^{\sqrt{d}} \rho(\mathcal{D}_m, \delta) d\delta \quad (30)$$

Remark 1: The value of $\varphi(\mathcal{D}_m)$ can indicate the quality of the local VDD. A more increased value of $\varphi(\mathcal{D}_m)$ implies improved data diversity, resulting in a better quality of the locally updated model. This is because greater spaces between traffic samples grasp more patterns during model training, enhancing the overall generalization of the global model. Let $\phi = \varphi_1, \dots, \varphi_M$ be the quality set of all SPS with M types, and all SPSs with $\varphi(\mathcal{D}_m) \in [\frac{m-1}{M}, \frac{m}{M}]$ hold a data of type m .

To solve Equation (15), the TPR can leverage each participant's previous performance, which might be time-consuming since assessing all upgraded models is a post-processing

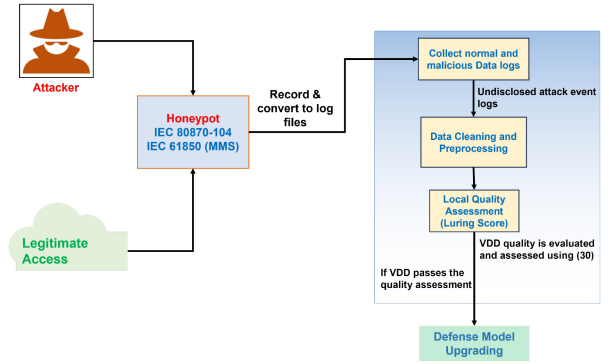


Fig. 3. Schematic illustrating how the luring score can be determined.

action. As a result, the quantity of $\varphi(\mathcal{D}_m)$ can be utilized to measure local VDD quality before a given SPS is selected. This approach allows the TPR to select high-quality data and avoid using models from participants with low-quality data, which can negatively impact the global model's performance. By employing this technique, the TPR can ensure that the final model is robust, even in the presence of malicious participants.

We can substitute $\pi_m = (V_m, R_m)$ in Section III by $\pi_m = (\Omega, R_m)$. Accordingly, we can further reformulate the optimization problem in (15) as follows:

$$\mathbf{P3.} \quad \max_{\Pi} : \sum_{m=1}^M S_z^m \varphi(\mathcal{D}_m) \quad (31)$$

such that

$$C1: \ln(\theta_1 R_1) - C_1 \geq 0, \quad (32)$$

$$C2: \ln(\theta_m R_m) - C_m \geq \ln(\theta_m R_{m-1}) - C_{m-1}, \forall m, \quad (33)$$

$$C3: \ln(\theta_m R_m) - C_m \leq \ln(\theta_m R_m) - C_{m+1}, \forall m, \quad (34)$$

$$C4: \sum_{m=1}^M R_m \leq B_z, \quad (35)$$

$$C5: T_m^{\text{total}} \leq T^{\text{max}} \quad \forall m \in \{1, 2, \dots, M\}, \quad (36)$$

$$C6: R_m > 0. \quad (37)$$

It is worth mentioning that the optimization problem (P3) considers the lure score to adaptively place honeypots in the network to maximize their luring potential. As in our proposed approach, the lure score can be calculated based on various factors such as network traffic patterns, historical attack vectors, and even machine learning models trained for this specific purpose. Fig. 3 illustrates how the luring score can be determined. The system ensures that honeypots are effectively deployed by considering the lure score, which will most likely capture meaningful data and contribute to better FL. Specifically, high-quality data, signifying that the user's honeypot has successfully deceived the attackers, is defined by a luring score. This leads to the development of more accurate and comprehensive models, which are essential for detecting and mitigating sophisticated cyber threats.

However, we note that the problem in (31) is still challenging to be solved directly due to the monotonic constraints in C1, C2, and C3. The authors in [34] proposed relaxing such

constraints, enabling the problem to be directly solvable as an optimization problem. However, in the context of FL, it is not practical to apply such solutions due to the following:

- The server cannot adequately analyze local updates even if all selected SPSs are trusted. The VDD quality metric can only be utilized as described in Section V-B only if all participants are completely trusted.
- Similar data may have varying contributions during the global training rounds, wherein it could initially make a higher contribution but gets reduced over time. For instance, the training process slows down if the model is closer to its stationary point, regardless of the associated cost or local data quality. Thus, the rewards should be given based on the value added to the global model every round, not as a fixed contract.
- The server should ensure fairness between the participants where the higher cost does not imply higher quality.

In the following section, we solve this problem iteratively by implementing a smooth FL process, including novel selection, aggregation, and averaging schemes. Then, we design a fair and efficient postprocessing rewards mechanism that fully ensures fairness so that the constraints (36–39) can be implicitly satisfied. The introduced solutions incorporate three phases; the first phase (i.e., the Preprocessing Phase) aims to solve an optimized relaxation problem to select the proper participants. The selected participants receive the global model, update it locally, and upload it to the TPR. In the second phase, the locally upgraded models by all participating SPSs are aggregated. The TPR evaluates the received models using two schemes (discussed below) and assigns a contribution rank for each model. In the third phase, the enhanced FedAvg based on adopted weights is applied under two scenarios when the participants are fully trusted and when some participants are malicious. The rewards are then given based on the contribution rank given to each participant (i.e., based on the uploaded model).

C. Quality-Assurance Model Averaging

The phases of the proposed solution are as follows. By resolving (31) during the initial phase (i.e., the preprocessing phase), it attempts to identify the appropriate SPSs. The identified participants will then receive the global model for local updating. The second phase combines the local updates from all SPSs to mitigate the challenges arising using the traditional FedAvg approach being assessed in two different settings.

1) *Trust-Based Model Averaging*: The TPR trusts all participants, and the weight of each participant is determined according to their reported data quality:

$$v_m = \frac{\varphi(\mathcal{D}_m)}{\sum_{m=1}^M \varphi(\mathcal{D}_m)} \quad (38)$$

Following that, the global model is generated and updated given by

$$V(\mathbf{z}) = \sum_{m=1}^M v_m \mathbf{V}_m. \quad (39)$$

In contrast to the conventional FedAvg, which utilizes the amount of data samples, this enhanced FedAvg does model averaging by weighting the local models according to the local data quality. Practically, weighting the model parameters according to the amount of data is futile since the data may contain redundancies and slightly affect the global model. Further details can be found in Section VI, where the conventional FedAvg approach demonstrates inferior performance.

D. Two-Steps Verification Mechanism

In this approach, the retailer initially selects the participating suppliers based on their local data quality. As explained in Section V-B, the higher the data quality, the better the model update. Thus, the server adopts the value of $\varphi(\mathcal{D}_m)$ to prioritize the suppliers and select the best set. However, the malicious supplier may claim higher data quality. Therefore, the server needs to apply a further verification step by testing the model update before confirming the supplier's reward. This can be done by estimating the Euclidean distance between two consecutive updates of the global model and the updated local model, respectively, or by having global data samples and feeding them into the uploaded models one by one before conducting the global averaging.

1) *Untrust-Based Model Averaging*: In this scenario, some participating SPSs might declare to have valuable data, yet the shared models may be perturbed or generated randomly. Consequently, a two-step verification is proposed, where the TPR initially assesses the obtained model employing the Euclidean distance or utilizing generalized test data evaluating every model independently. Assuming the latter is utilized, the adjusted averaging weights can be determined using the formula as follows:

$$w_m = \frac{G(v_m)}{\sum_{m=1}^M G(v_m)} \quad (40)$$

Following that, the model is generated and updated based on (39).

E. Incentive Reward design

We propose applying the Soft-max function to determine the incentives provided to the selected SPSs after computing the authentic average weights employing one of the methods outlined in the preceding section. This will guarantee equality according to the contribution provided by each SPS. It can be expressed as follows:

$$w_m = \frac{e^{v_m}}{\sum_{m=1}^M e^{v_m}}. \quad (41)$$

It is important to note that for each round z , the total rewards will be less than the budget assigned. As a result, the participants receive the earned rewards from:

$$R_m = w_m B_z. \quad (42)$$

We may infer from (41) and (42) that the obtained rewards ensure equality for all SPSs participants and completely meet the assigned budget. It is worth noting that our approach considers the heterogeneity of threats and their distribution

Algorithm 1: FedPot Framework

Input: All available SPSs M
Output: Upgraded Security Model V

- 1 **Initialize:** starting global model V_0 , local epochs ε , step size η , rounds Z , and assigned budget for every global round;
- 2 TPR collects previous information (e.g., $\varphi(\mathcal{D}_m)$), from existing SPSs M ;
- 3 **for** $z = 1$ **to** Z **do**
- 4 The TPR establishes the time limit (i.e., deadline);
- 5 TPR addresses the optimization problem, **P3**, to choose the best entities for participating in the model update;
- 6 TPR shares the model V_{z-1} to chosen SPSs;
- 7 **for** Each SPS $m \in M$ **synchronously do**
- 8 SPS m gets V_{z-1} ;
- 9 SPS m updates m employing its VDD obtained data for E epochs;
- 10 SPS m uploads V_m to TPR;
- 11 The TPR gathers all submitted updates from participating SPSs;
- 12 **if** Every participant is trusted **then**
- 13 TPR employs Equation (38) to recompute the weight associated with each update.
- 14 **else**
- 15 TPR assesses each model using the generalization test data;
- 16 TPR utilizes Equation (40) to recompute the weight for every update
- 17 TPR utilizes equation (39) to generate a modified global model.

implying that most SPSs, over a reasonable time frame, are exposed to some kind of malicious activity. It might be accurate that at certain times, specific SPS face a higher volume or intensity of attacks, but when viewed over an extended period, the distribution becomes relatively even. We have designed the reward mechanism to factor in not just the volume but also the quality of the data. This way, even if an SPS does not face a high number of attacks, the uniqueness or novelty of a single quality data point they contribute could be of immense value, ensuring they are adequately rewarded. Alg. 1 outlines the overall process of the suggested approach. In Alg. 1, the TPR starts by initializing the model and the hyper-parameters (step 1), then gathering prior information from all SPSs (step 2). From steps 3–17, the TPR runs the FL algorithm by determining the deadline for each round (step 4) and solving the relaxed **P3** (step 5) to select the best participants while accounting for the stated data quality. The global model is then sent (step 6), and all selected participants undertake local training and submit all local models once finished (steps 7–10). The TPR aggregates all models (step 11) before deciding on one of the proposed schemes. If the TPR can confirm the trustworthiness of the participating SPSs, the solution in Section V-C is applied (steps 12 and 13), and the solution in Section V-D is applied (steps 15 and 16) if TPR can not. Finally, as explained in Section V-E, the rewards are fairly given to the participants efficiently.

F. Architectural Mapping of the Proposed Solution

It is worth noting that our control framework is designed to emulate the hierarchical architecture commonly observed in modern SG systems. It comprises three main layers: the Data Acquisition Layer, the Control Layer, and the Decision Layer. **Data Acquisition Layer:** This is the base layer where all sensors and IoT devices (as represented by our use of the

N-BaIoT dataset in Section VI) are located. These devices are responsible for collecting real-time information such as voltage, current, and frequency from different points in the grid. **Control Layer:** where the real-time data is analyzed and control signals are generated. This layer integrates the security protocols as studied through the IEC 104 dataset in section VI-C. The layer includes controllers like Remote Terminal Units (RTUs) and programmable logic controllers (PLCs), which interact directly with the devices in the Data Acquisition Layer. The control layer continuously monitors and analyzes real-time data, identifying potential cyber threats or operational anomalies. This layer generates signals, executes responses to mitigate threats, and integrates various security protocols. A key feature is the strategic deployment of honeypots within the network. These honeypots serve as decoy systems designed to attract cyber attackers. This allows for an in-depth analysis of attack strategies, significantly enhancing the grid's cyber resilience and providing valuable insights into potential vulnerabilities. **Decision Layer:** This is the topmost layer, consisting of control centers or cloud-based systems where higher-level decision-making processes occur. Here, our proposed anomaly detection algorithms and security measures are implemented (i.e., ML-based detector). To generate realistic grid scenarios, we employed both the N-BaIoT and IEC 104 datasets, creating a diverse set of operating conditions and cyber-attack vectors. Our framework is also designed to be scalable and robust, which allows for the integration of additional sensors and control units as required. The proposed incentive-based model is implemented at the Decision Layer, ensuring that it benefits from the real-time data collected and analyzed at the lower layers and addressing the free-rider problem. This facilitates more effective and timely decision-making. It is worth noting that in this layer of the context of cybersecurity, test datasets cannot remain static. When a new type of attack is detected by a honeypot, it may initially be evaluated through heuristics, expert rules, and anomaly detection methods for interim validation. Contextual information provided by the SPS also helps the TPR to assess the contribution's potential impact. To corroborate new threats, we propose sharing them (anonymously) with a subset of trusted SPS for additional verification. As these new types of attacks are validated, they will be incorporated into future iterations of the test data, ensuring a relevant benchmark for subsequent evaluations.

G. Theoretical Analysis of Robustness Against Malicious SPSs

To validate the resilience of our Quality-Assurance Model Averaging and Two-step Verification Mechanism against malicious activity, we introduce some theoretical metrics and analysis. Let $\Delta V(z)$ denote the deviation of the global model $V(z)$ under the Trust-based Model Averaging scheme given by (39):

$$\Delta V(z) = \|V(z) - V^*\| \quad (43)$$

Here, V^* represents the optimal global model that would have been achieved without any malicious activity. We aim to bound

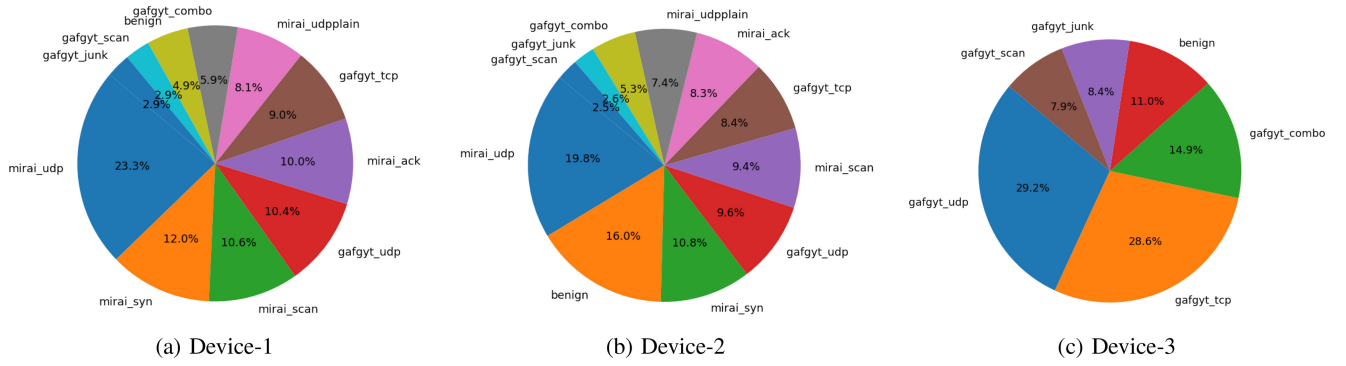


Fig. 4. Label distribution among the logs of the first three devices.

$\Delta V(z)$:

$$\Delta V(z) \leq f(\varphi(\mathcal{D}_m), M, v_m, \dots) \quad (44)$$

Here, f is a function that encapsulates the contributions of local data quality $\varphi(\mathcal{D}_m)$, the total number of participants M , and their respective weights v_m , among other parameters. Similarly, we define $\Delta W(z)$ as the deviation under the Untrust-based Model Averaging scheme given by (40).

$$\Delta W(z) = \|V(z) - V^*\| \quad (45)$$

We aim to bound $\Delta W(z)$ as follows:

$$\Delta W(z) \leq g(G(v_m), M, w_m, \dots) \quad (46)$$

Here, g is a function capturing the effects of the generalized test $G(v_m)$, total number of participants M , and the adjusted weights w_m , among others. In essence, the bounded nature of $\Delta V(z)$ and $\Delta W(z)$ implies that our proposed schemes are robust against a variable number of malicious SPSs, maintaining the global model's efficacy.

VI. PERFORMANCE EVALUATION

In this section, we establish an experimental setting to evaluate the efficacy of our proposed methods for improving the security defense model and allocating fair incentives.

A. Experimental Setup

We employ the N-BaIoT [35] as well as IEC 104 and IEC MMS datasets [36]. N-BaIoT was initially designed to cover the security aspects of IoT devices. This dataset is crucial for modern SG systems increasingly relying on IoT sensors and actuators for efficient and intelligent grid management. It includes data from benign operations as well as from a range of attacks like Mirai and Gafgyt (Bashlite). Mirai attacks focus on different flooding techniques using ACK, SYN, UDP, and other protocols while performing device vulnerability scans. Gafgyt attacks, similarly, engage in device scanning and deploy a mix of attack strategies, including TCP and UDP flooding. As in Fig. 4 for the first three devices, it consists of eleven classes, one benign, while the remaining ten represent adversarial classes (i.e., indicating malicious data), with each sample including 115 attributes. These datasets, in particular, include DDoS, which causes networks to become

overloaded with traffic; bot attacks that take advantage of device flaws; man-in-the-middle attacks that intercept communications; SQL injection that manipulates databases; and XSS which injects malicious scripts. Firmware flashing, physical tampering, and side-channel attacks also represent advanced threats. In SG, DDoS attacks can disrupt grid communications, Bot attacks may compromise grid control systems, and MitM attacks pose risks to data integrity. SQL injection and XSS are relevant where SGs use Web interfaces or databases. Firmware flashing and physical Tampering highlight the need to secure grid hardware, while side-channel attacks demonstrate the importance of protecting against indirect data leaks.

We compare our proposed method with the traditional FedAvg technique, where the server allocates weights to the local model updates based on the amount of traffic data samples they include. Furthermore, we evaluate the incentives-based algorithm against a customized-based algorithm. We employ a supervised deep neural network (DNN) model consisting of 115 input layers followed by 115, 62, and 32 hidden layers in which $\frac{1}{2}$, $\frac{1}{4}$, and $\frac{1}{8}$ inputs follow each hidden layer, respectively. The cross-entropy is used as a loss function. We initially consider the N-BaIoT dataset's initial device count, consisting of nine devices exhibiting non-i.i.d. characteristics. Similar parameters are adopted for all experiments. There are eight SPSs as participants in the FL configurations, with each SPS possessing one out of the nine devices present in the N-BaIoT datasets, including both benign and adversarial samples. One device is solely utilized to evaluate the overall model developed by each of the other eight participants. We conduct each experiment five times to ensure the accuracy of the findings. Then, 50 devices are added to the data distribution. Every device uses a 10 local epoch and a 32 local batch size. We employ an adjustable learning rate with a default value of 0.01.

We assess the efficiency of our proposed scheme using the following metrics accuracy, loss, true positive rate (TPRate), false positive rate (FPR), and F1-measure given by: $TPRate = \frac{TP}{TP+FN}$, $TNR = \frac{TN}{TN+FP}$, and $F1-score = \frac{2 * Precision * Recall}{Precision + Recall}$. To assess the model's efficacy, we utilize the testing accuracy, which measures the proportion of accurately identified used incursions as a performance indicator. Finally, we use rewards distribution fairness to showcase the efficiency of the proposed rewards allocation framework.

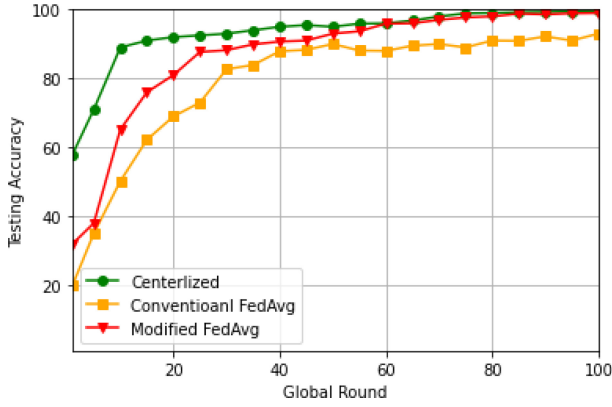


Fig. 5. Comparative Analysis of Test Accuracy: Enhanced FedAvg, Conventional FedAvg, and Centralized-Based Algorithms.

B. Numerical Results on BalIoT Dataset

In this section, we perform our numerical analysis on IDD and non-i.i.d. assumptions.

1) *The IID Setting*: Fig. 5 provides an overview of the trust-based model averaging framework performance under several distinct settings: complete data offloading to TPR (centralized training), the conventional FedAvg, and the proposed enhanced-FedAvg (FedAvg based on data quality). The results of the proposed algorithm demonstrate that it outperforms the conventional FL approach, which can be attributed to the effect of assigning model weights according to the local data quality instead of the data size as in conventional FedAvg. Furthermore, the enhanced FedAvg scheme can attain almost traditional centralized performance. In contrast, the traditional FedAvg algorithm needs more rounds to obtain the required accuracy, which increases the cost, leading to surpassing the budget assigned.

Fig. 6(a) shows the impact of the untrust-based model averaging framework under two scenarios: when all participants are completely trusted and when some are malicious. Our observations reveal that depending solely on the data quality asserted by SPSs, the presence of corrupted models leads to subpar models. This outcome arises due to malicious participants who assert high-quality data yet disperse it arbitrarily or manipulate local models. The suggested untrusted model averaging approach may efficiently remove detrimental models by assigning weights to the shared models according to the generalization test data. This causes the effects of adversarial or disturbed models to vanish during the averaging phase or at least reduces their negative impact on the designed global model utilized for NIDS. Contrarily, malicious participants may significantly reduce the defensive model's effectiveness, despite the data quality they claimed throughout the selection process. We repeat similar experiments while splitting the data between 50 and 100 SPSs to showcase the effectiveness of the proposed approaches.

As shown in Fig. 6(b) and 6 (c), we use similar settings but distribute the data across 50 and 100 SPSs and keep the labels in an i.i.d. fashion. We follow the original data distribution, where the labels across the SPSs follow the same distribution. We observe that even if the data is i.i.d., the presence of some

TABLE I
COMPARATIVE RESULTS FOR TPRATE AND TNR OF CONVENTIONAL FEDAVG AND PROPOSED SOLUTIONS UNDER ATTACK AND NON-ATTACK SCENARIOS (I.I.D. DATA DISTRIBUTION, N-BalIoT DATASET)

Alg.	C on. FedAvg		P -Trusted		P -Untrusted	
Malicious SPSs	No	Yes	No	Yes	No	Yes
TPRate	89.7	52.9	92.95	76.3	95	93
TNR	62.1	30.02	85	61.9	90.85	88
Fairness	50.01	31	93.9	55	95.1	94.7

malicious participants affects the performance of the models, especially as the number of SPSs increases. Nevertheless, the untrust-based averaging scheme still performs better than the trust-based scheme. Yet, the accuracy is slightly decreased as the number of SPSs increases.

Furthermore, we assess the efficacy of the proposed approaches regarding the fairness of the reward using the true positive rate (TPRate) and the true negative rate (TNR), critical performance indicators for the rewards allocation and detection rate, and in the security defense model. Table I shows that, whether malicious participants are present or not, both proposed approaches exceed the traditional FedAvg. Nonetheless, in the presence of malicious participants, the proposed trust-based model averaging (PTrusted) performance exhibits a considerable decline in TPRate, TNR, and fairness of rewards. This degradation occurs as we solely depend on the claimed data quality. On the other hand, the proposed untrust-based model averaging ensures adequate outcomes despite malicious participants. This is achieved through a two-step verification process that excludes altered models and accordingly assigns rewards, thereby achieving a high level of fairness. Meanwhile, FedAvg distributes the incentives randomly depending on the data size.

2) *The Non-IID setting*: For the non-i.i.d. setting, we assume that each SPS holds only a maximum of 2 types of attacks (i.e., Mirai UDP and combo). We consider three scenarios regarding the number of devices: 9, 50, and 100. We assign only a maximum of two classes for each SPS to ensure that the datasets amongst SPSs are non-i.i.d. The default data distribution is considered non-i.i.d. However, as illustrated in Fig. 4, the labels' distribution across all devices is almost identical, even for those holding only 6 classes. Hence, in this work, we consider a realistic non-i.i.d. distribution to evaluate the performance in the worst situations.

As shown in Fig. 7(a), we initially run the first scenario when the number of devices is only 9. One can see that the performance in general drops by approximately 15% for both proposed averaging schemes. However, when some participants maliciously claimed high-quality local logs during the selection phase and uploaded perturbed models, the performance of the first proposed scheme (i.e., trust-based model averaging) dramatically dropped by almost 50% to reach 46% accuracy. This is due to adopting inaccurate weights during the evaluation phase, thereby impacting the performance of the global model. In contrast, the second proposed scheme (i.e., untrust-based model averaging) shows

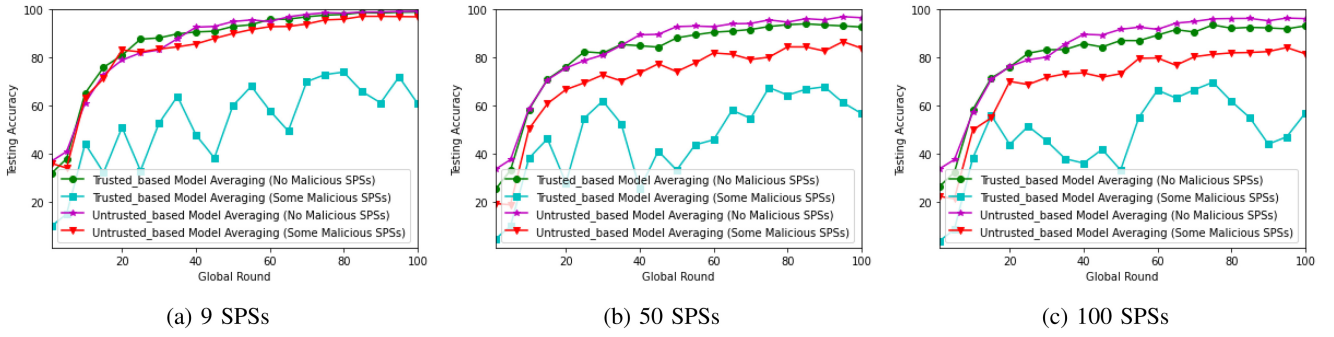


Fig. 6. Test Accuracy Comparison of Proposed Schemes in the Presence of Malicious SPSs Claiming High-Quality Data While Uploading Defective Models (i.i.d. Data Distribution, N-BaIoT dataset).

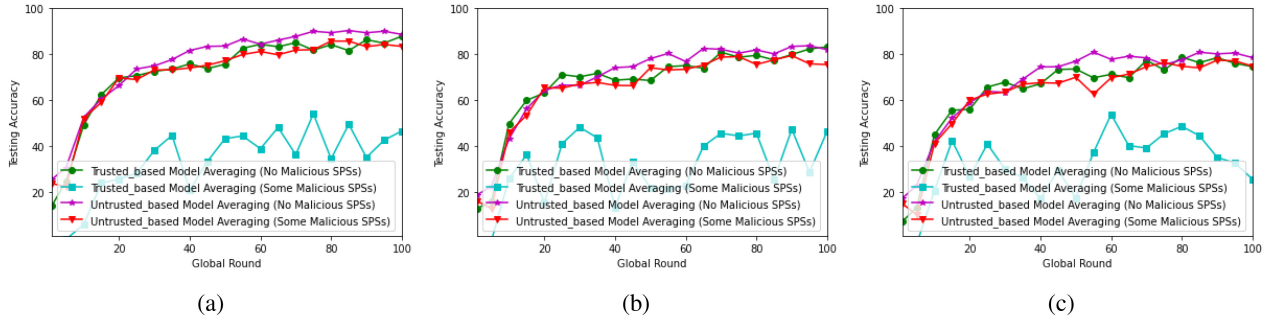


Fig. 7. Test Accuracy Comparison of Proposed Schemes in the Presence of Malicious SPSs Claiming High-Quality Data While Uploading Defective Models (non-I.I.D. Data Distribution, N-BaIoT dataset).

TABLE II
COMPARATIVE RESULTS FOR TPRATE AND TNR OF CONVENTIONAL FEDAVG AND PROPOSED SOLUTIONS UNDER ATTACK AND NON-ATTACK SCENARIOS (NON-I.I.D. DATA DISTRIBUTION, N-BAIoT DATASET)

Alg.	C on. FedAvg		P -Trusted		P -Untrusted	
Malicious SPSs	No	Yes	No	Yes	No	Yes
TPRate	78.3	31.6	84.5	34.2	85.2	81.7
TNR	56.4	31.4	82.6	46.2	82.3	79.1
Fairness	51.2	19.7	97.1	41.2	98	96.9

immunity against the perturbed models even when the data is non-i.i.d.

In Fig. 7(b) and 7(c), we repeat the same experiments, distributing the data across 50 and 100 SPSs, respectively. From a security perspective, the performance in both figures almost matches the performance in Fig. 7(a), where the second scheme outperforms the first averaging scheme when attacks are present. However, the convergence becomes slightly slower as the number of SPSs grows. In the case of massively distributed data due to the increased number of SPSs, more rounds are required to capture patterns from all available SPSs.

We also investigate the effect of all proposed schemes, including the reward mechanisms, on the overall performance in terms of detection rate (i.e., TPRate and TNR) and the fairness of reward distribution. From Table II, it can be shown that proposed averaging methods significantly surpass the conventional FedAvg, whether or not adversarial participants are present. However, when particular participants

are adversarial, the proposed trust-based model averaging (P-Trusted) suffers drastically in terms of TPRate, TNR, and reward fairness, which is unsurprising as these are heavily based on the data quality. In contrast, the proposed untrust-based model averaging successfully filters the contaminated uploaded models using a two-stage verification process. It distributes the rewards appropriately, achieving a high degree of fairness despite the presence of adversarial participants. FedAvg, on the other hand, distributes its incentives randomly according to the amount of data gathered. Nevertheless, all schemes are affected by the data distribution amongst SPSs, as we note when comparing the i.i.d. to non-i.i.d. results.

C. Numerical Results on IEC 104 Dataset

The IEC 104 Dataset was released by Brno University in March 2022 and represents a significant resource in the field of SG security [36]. It comprises IEC 104 and IEC MMS headers and is primarily designed for anomaly detection and security monitoring. The dataset captures a broad spectrum of attack scenarios across different folders. For instance, the but-iec104-i folder contains various attacks, including Denial of Service (DoS), Injection, and Man-in-the-Middle (MITM) attacks. These offer a comprehensive view of the threat landscape in the Industrial Control Systems (ICS) communications. Another folder, vrt-iec104, presents a different set of intriguing attacks, such as value change and masquerading attacks. It is worth mentioning that The IEC 60870-104 (IEC 104) and IEC 61850 (MMS) datasets, produced by the “Security monitoring of communication in the smart grid (Bonnet)” project at the Brno University of

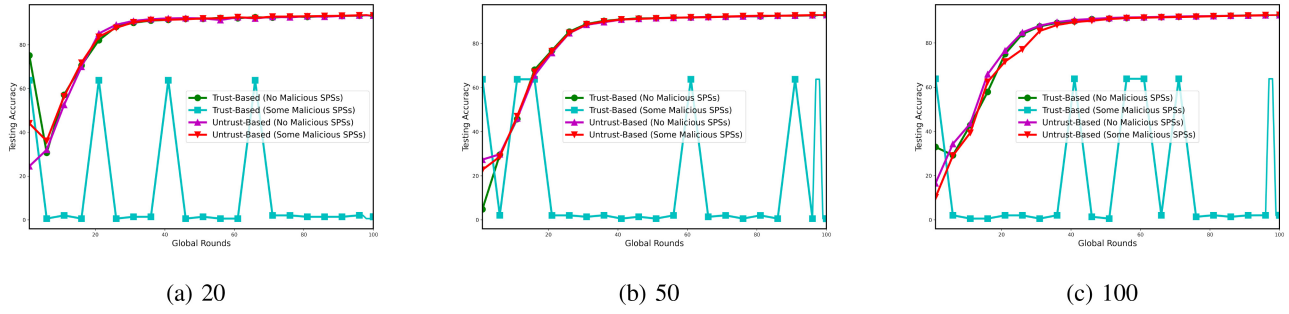


Fig. 8. Test Accuracy Comparison of Proposed Schemes in the Presence of Malicious SPSs Claiming High-Quality Data While Uploading Defective Models (I.I.D. Data Distribution, IEC 104 dataset).

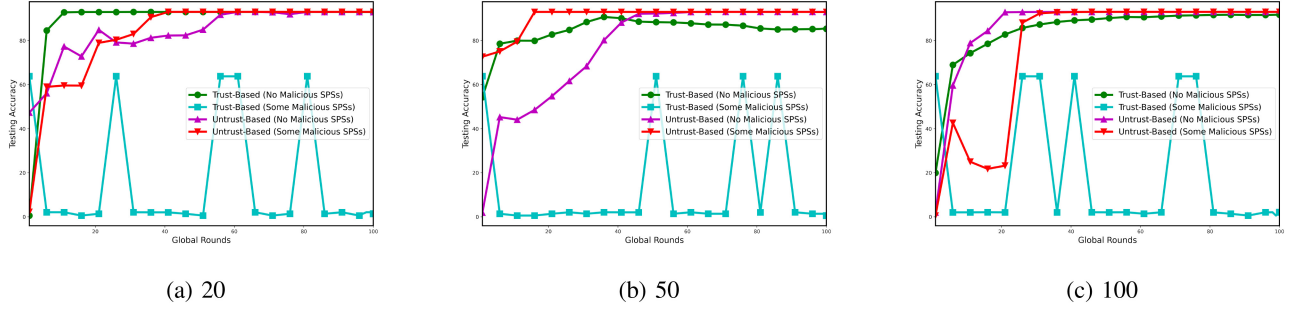


Fig. 9. Test Accuracy Comparison of Proposed Schemes in the Presence of Malicious SPSs Claiming High-Quality Data While Uploading Defective Models (non-I.I.D. Data Distribution, IEC 104 dataset).

Technology, Czech Republic (2019-2022), include CSV traces from PCAP files [36]. These datasets, derived from both real device observations and virtual application monitoring, provide a comprehensive view of normal and attack traffic patterns within smart grid environments. This dual approach, blending real-world operational traffic with simulated attack scenarios, ensures the datasets support robust model training and validation, preparing the models to detect and mitigate both existing and emerging threats effectively. Despite its richness, the dataset was originally unlabeled and required substantial preprocessing. This step was critical to ensure that our subsequent analyses were based on accurate, well-defined data. The dataset is organized into multiple folders, each containing data related to either IEC 104 or IEC MMS headers. Each folder has a readme.txt file that provides valuable information, including data types and timestamps related to the attacks. This level of detail enabled us to better understand the structure and implications of the data. We performed preprocessing steps through a multi-stage process to adequately prepare the data for our experimental framework. This involved labeling the samples into categories such as benign, switching, scanning, or communication interruption; cleaning the data to remove any irrelevant or inaccurate information, and finally transforming it into a format suitable for our machine learning algorithms. The encoding was significant for nominal or categorical data. The following section focuses on the numerical analysis conducted using the IEC 104 dataset. We examine performance under two conditions: the i.i.d. and non-i.i.d. settings.

1) *The IID Setting:* We consider three scenarios featuring varying numbers of substations: 20, 50, and 100. Fig. 8(a)–8(c) display the performance of our framework, which includes

both trust-based and untrust-based schemes when the data is IID (i.e., all clients have the same data distribution). Our results clearly demonstrate the superiority of the enhanced FedAvg, which achieves performance levels nearly identical to centralized models. This can be attributed to our focus on local data quality for model weight assignment as opposed to merely considering data volume, as in traditional FedAvg. This advantage holds even when malicious participants are involved. Performance significantly drops and fluctuates when only the trust-based scheme is employed. This drop is due to the exclusive reliance on local data quality. However, our untrust-based averaging scheme effectively mitigates this by leveraging generalization test data to allocate weights, neutralizing the impact of corrupted local models.

2) *The Non-IID Setting:* In this setting, we assume that each SPS is limited to specific types of IEC 104 traffic patterns. Fig. 9(a)–9(c) show that our proposed untrust-based averaging scheme performs exceptionally well, even in the presence of malicious activity. Trust-based model averaging struggles in this context, experiencing a significant decline in accuracy due to its reliance on claimed data quality. Furthermore, Table III reveals that our approaches outperform traditional FedAvg in critical security metrics such as TPRate and TNR, especially when adversarial participants are present. The trust-based model sees a decline in these metrics due to its dependence on claimed data quality. In contrast, our untrust-based model averaging maintains robust performance, demonstrating its value in real-world scenarios where malicious activity is a concern. Notably, our proposed approaches exhibit superior performance in non-IID settings. This is due to our emphasis on evaluating the quality of returned contributions rather

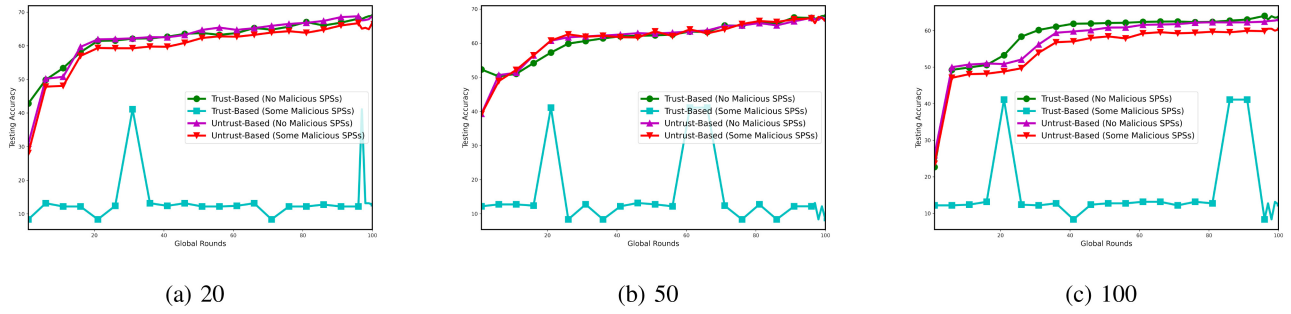


Fig. 10. Test Accuracy Comparison of Proposed Schemes in the Presence of Malicious SPSs Claiming High-Quality Data While Uploading Defective Models (non-I.I.D. Data Distribution, IEC 61850 (MMS) dataset).

TABLE III

COMPARATIVE RESULTS FOR TPRATE AND TNR OF CONVENTIONAL FEDAVG AND PROPOSED SOLUTIONS UNDER ATTACK AND NON-ATTACK SCENARIOS (NON-I.I.D DATA DISTRIBUTION AND IEC 104 DATASET)

Alg.	C on. FedAvg		P -Trusted		P -Untrusted	
Malicious SPSs	No	Yes	No	Yes	No	Yes
TPRate	81.2	28.2	86	23.8	92.7	87.3
TNR	63.1	25.1	86.2	37.4	85.3	83.6
Fairness	69.8	14.3	98.6	32.7	97.8	94.6

than just data quantity, leading to a more generalizable global model. In summary, our numerical evaluation of the IEC 104 dataset validates the effectiveness of our proposed methodologies, particularly in non-i.i.d settings and when malicious participants are involved. The insights gained from this analysis will be crucial for refining and optimizing our models for intrusion detection in ICS.

D. Verification With IEC 61850 Manufacturing Message Specification (MMS) Dataset

To further verify our results, we carry out experiments using the IEC 61850 (MMS) dataset. IEC 61850 (MMS) is extensively employed in electric utility companies, predominantly for substation automation and ensuring interoperability between different manufacturers' systems. This global standard is important for streamlining the communication infrastructure of electrical substations and facilitating the integration, operation, and maintenance of diverse devices and systems in SG. Its usage spans various applications, including real-time monitoring, control of substation components, and ensuring robust, reliable data exchange. The dataset used includes both benign and malicious data samples. Remarkably, the performance trends observed with this dataset closely mirror those identified using the IEC 104 protocol data. Specifically, under non-IID settings as a most challenging scenario, our enhanced FedAvg framework, which includes both trust-based and untrust-based schemes, continued to demonstrate superior performance, closely approximating that of centralized models. This consistency highlights the robustness of our approach, particularly our novel method of weighting model updates based on the quality of local data rather than its volume. Including the IEC 61850 dataset not only verifies our initial findings but also broadens the

applicability of our framework. It shows that our untrust-based averaging scheme, which assesses the quality of contributions through generalization on test data, can effectively neutralize the influence of corrupted local models, ensuring stable and reliable model performance even in the face of malicious activity. This result demonstrates the generalizability and robustness of our approach across different SG communication protocols, further validating the effectiveness of our method in enhancing security and reliability in real-world, heterogeneous SG network environments.

VII. CONCLUSION

In this paper, we introduced FedPot, a novel quality assurance honeypot-based FL framework designed for network security in SG. FedPot incorporates a novel, efficient, and resilient aggregation and averaging schemes coupled with a fair rewards mechanism. We presented novel schemes for local data quality, participant selection, and global model upgrading using the N-BaIoT, IEC 104, and IEC MMS datasets. In FedPot, the TPR addresses a convex optimization problem, prioritizing data quality over data size. Each SPS optimizes the global model with its honeypot logs and transmits the model updates back to the TPR. Subsequently, the TPR enhances the defensive model using the approaches proposed in this study. To mitigate the free-rider issue prevalent in AMI networks within the FL framework, we proposed a new metric to gauge local data quality and contributions, eliminating the need to rely on data size. We also devised a two-step verification process to address the challenge of adversaries or underperforming SPSs. Additionally, we introduced an improved FedAvg scheme for local model aggregations. The results obtained from extensive simulations with realistic log data attest to the effectiveness of our proposed scheme, which outperforms current state-of-the-art techniques. As a direction for future research, investigating the real-time implementation and assessment of FedPot in streaming, diverse, and larger-scale environments would be insightful. Lastly, adapting the FedPot framework to address other cybersecurity threats within various IIoT applications could also be advantageous.

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